

Dheeraj Dhillon

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EDUCATION

University of Wisconsin-Madison

Aug 2025 - May 2027

Master of Science (M.S.), Computer Science

Coursework: Reinforcement Learning, Machine Learning, Game Theory, Operating Systems

Indian Institute of Technology (IIT) Roorkee

Jul 2019 - May 2023

Bachelor of Technology (B.Tech.), Electronics and Communication Engineering

Cumulative GPA: 8.491/10.0

SKILLS

Programming Languages: Python, C++, C, SQL

Frameworks & Tools: PyTorch, NumPy, Databricks, Spark, Gymnasium, ALE, MuJoCo, TRL, Hugging Face Transformers, scikit-learn, statsmodels, spaCy, NLTK

WORK EXPERIENCE

Gartner Inc., Associate Data Scientist, Gurgaon

Mar 2024 - Aug 2025

- Built and productionized a 3-level topic modeling pipeline in Databricks/Spark over 20k user queries and 30k Gartner research documents using all-MiniLM-L6-v2 embeddings and Louvain clustering to surface demand-sensing insights.
- Developed Implied Value Rating (IVR), a document-level metric estimating the model-predicted change in retention probability associated with readership, used to inform search reranking toward high-impact content.
- Calculated IVR for 21k+ documents across 17 contract types and 100k licensed users by fitting one maximum-likelihood logistic regression per document and measuring the predicted retention delta under read versus not-read settings; parallelized model fitting with Spark RDDs across 24 workers.
- Explained drivers of document performance by regressing IVR against topic profiles, uniqueness, and engineered semantic, discourse, and NER features; identified 50+ significant attributes to inform author content planning and achieved adjusted R-squared of 0.62 on 4k IT End User documents.

HiLabs Inc., Data Scientist, Bangalore

Jul 2023 - Mar 2024

- Built a clinical information extraction pipeline over 20k OCR-extracted EMR notes, using CRF-derived entity spans and lexico-syntactic dependency patterns to generate Subject-Predicate-Object (SPO) triples for downstream clinical data standardization, search, and interoperability.
- Developed a linear-chain Conditional Random Field (CRF) sequence tagger with BIO encoding for 30 clinical entity types, achieving 82% strict span-level F1 on a held-out annotated EMR set.
- Evaluated contextual and linear bandit strategies for personalized warfarin dosage selection under sequential feedback, comparing Epsilon-Greedy, Upper Confidence Bound (UCB), Thompson Sampling, and LinUCB.
- Benchmarked online decision accuracy at 68% on dose selection, highlighting the trade-off between exploration and patient-specific treatment optimization.

Microsoft R&D, Data and Applied Scientist Intern, Hyderabad

May - Jul 2022

- Deployed a BM25-based message recommender for Microsoft Teams Search and integrated LDA-based query expansion, increasing NDCG@10 from 31% to 40% on an 8,000-message dataset.
- Utilized KeyBERT and TextRank to perform keyphrase extraction (KPE), and designed a semantic matching evaluator using Sentence Transformers with an adjustable cosine similarity threshold.
- KeyBERT produced the strongest KPE result on the Teams dataset, reaching F1@5 of 55% under semantic cosine-similarity matching, compared with 43% under exact matching.

PUBLICATIONS

InfraNet: An Ensemble Approach for Real-time Wildlife Detection using Infrared Thermal Imaging, IEEE AVSS Aug 2025

PROJECTS

Reinforcement Learning from Human Feedback (RLHF) for LLM Alignment

May 2026

- Built a three-stage LLM alignment pipeline for Qwen2.5-0.5B-Instruct using 38K+ HelpSteer3 examples, spanning LoRA-based supervised fine-tuning (SFT), reward modeling, and KL-regularized Proximal Policy Optimization (PPO).
- Trained a Bradley-Terry reward model achieving 65% validation accuracy. Evaluated Base, SFT, and PPO policies on 2,017 held-out prompts using pairwise reward scoring, length controls, repetition diagnostics, and qualitative analysis.
- Deployed a live interactive response explorer with domain filters for General, STEM, Code, and Multilingual prompts to compare Base and PPO generations and identify reward-model mismatches and alignment regressions.

Trust Region Policy Optimization and Policy Gradient Methods, UW-Madison

Mar-Apr 2026

- Implemented Trust Region Policy Optimization (TRPO) from scratch in PyTorch, from trajectory collection and advantage estimation to KL-constrained policy optimization using conjugate gradient updates, Fisher-vector products, and line search.
- Implemented PPO with both clipped and KL-penalized objectives, and benchmarked TRPO and PPO on 3 MuJoCo locomotion tasks and 7 Atari games using Gaussian MLP policies for control and categorical CNN policies for Atari.
- Analyzed monotonic policy improvement guarantees in CPI and TRPO through surrogate performance objectives, clarifying how mixture-update constraints in CPI and KL-divergence constraints in TRPO stabilize policy updates.